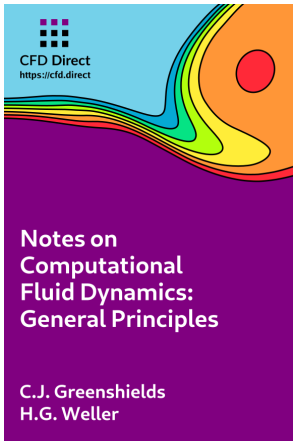
Notes on Computational Fluid Dynamics: General Principles



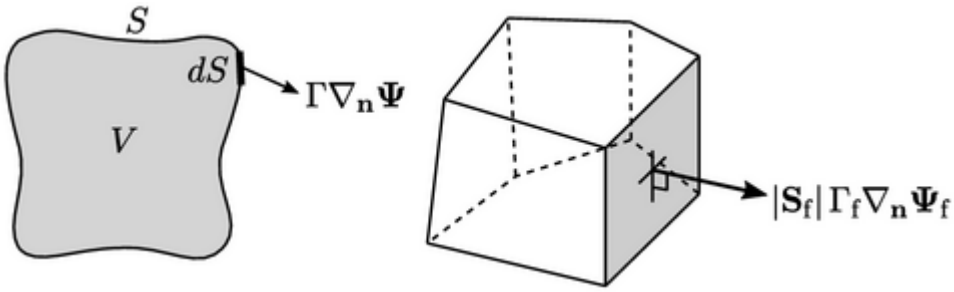
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3.7 Laplacian discretisation

Let us first describe the discretisation of the Laplacian term for diffusion, introduced in Sec. 2.14.



Following the finite volume principles described in Sec. 3.1, the discretisation approximates the surface integral by a summation over faces by

$$\nabla\cdot(\Gamma\nabla\Psi) = \lim_{\Delta V\to 0}\frac{1}{\Delta V}\int_S (dS\,\Gamma\nabla_n\Psi) \rightarrow \frac{1}{V}\sum_f |\mathbf{S}_f|\,\Gamma_f\nabla_n\Psi_f. \tag{3.2}$$

The $\sum_f$ in Eq. (3.2) can be viewed as the summation over faces of a single cell. Applying the summation to all cells provides the contribution to coefficients $[\mathbf{A}]$ and $[\mathbf{b}]$ of a matrix equation.

The mesh data V and $\mathbf{S}_f$ are calculated according to Sec. 3.3 so the remaining properties to be determined are:

- the diffusivity at faces Γ_f ;
- the surface normal gradient at faces $\nabla_n\Psi_f$.

Fields Γ and Ψ are associated with cells, so numerical schemes are required to evaluate properties at faces. We will first describe

interpolation for Γ_f, for which the linear scheme is generally used. The surface normal gradient $\nabla_n \Psi_f$ is discussed in Sec. 3.8.

Interpolation from cells to faces

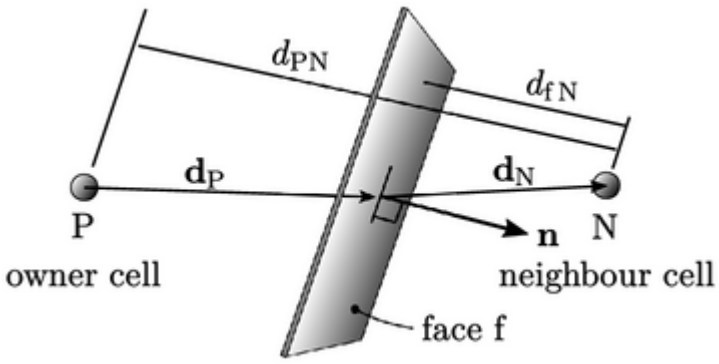
Mapping data between different locations is a common practice in numerics. Since the finite volume method is concerned with fluxes at faces, the principal mapping procedure is *interpolation* from cells to faces. Since other interpolations are much less common, it can be assumed that the term "interpolation" means from cell to face unless stated otherwise.

For irregular polyhedral meshes, interpolation is generalised by defining a *weights* field w for each face according to

$$\Psi_f = w\,\Psi_P + (1-w)\,\Psi_N, \tag{3.3}$$

where Ψ_f is the interpolated face field. The subscripts P and N indicate values at owner and neighbour cells, respectively.

Linear interpolation



The **linear** interpolation scheme sets Ψ_f according to a linear variation between cells values Ψ_P and Ψ_N. The weights can then be calculated based on distances from the face centre to adjacent cell centres, in the direction normal to the face, by

$$w = \frac{d_{fN}}{d_{PN}} = \frac{\mathbf{n}\cdot\mathbf{d}_N}{\mathbf{n}\cdot(\mathbf{d}_P + \mathbf{d}_N)}. \tag{3.4}$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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